



Azure Front Door

What is Azure Front Door?

Azure Front Door is a scalable and secure **global application delivery network (ADN)** service that helps optimize the performance, security, and availability of web applications. It operates at **Layer 7 (HTTP/HTTPS)** of the OSI model and provides **intelligent traffic routing, caching, security features, and global load balancing** for applications running on Azure or other cloud platforms.

Key Use Cases of Azure Front Door

Azure Front Door is beneficial in various scenarios, including:

1. Global Load Balancing

- Distributes traffic across multiple backend regions, ensuring **high availability and fault tolerance**.
- Routes traffic based on latency, priority, or geographic location.

2. Accelerating Web Applications

- Uses **Anycast DNS routing** and **Microsoft's global network** to optimize request handling.
- Improves **website performance** by reducing latency.

3. DDoS Protection & Web Application Security

- Integrates with **Azure Web Application Firewall (WAF)** to protect against common cyber threats (SQL injection, cross-site scripting, etc.).
- **Rate limiting** and **bot protection** features enhance security.

4. SSL/TLS Offloading

- Terminates SSL connections at the edge, reducing the load on backend servers.
- Provides **automatic certificate management** for secure communication.

5. Multi-Cloud and Hybrid Deployments

- Can distribute traffic between **on-premises, multi-cloud, and Azure-based** applications.
- Enhances redundancy and resilience.

6. Custom Traffic Routing

- Routes requests based on **URL patterns, query strings, HTTP headers, and other rules**.
- Supports **A/B testing** and **blue-green deployments**.

7. Content Caching and Compression

- Uses **edge caching** to store static content closer to users, reducing latency.
- Compresses responses for faster delivery.

Benefits of Azure Front Door

1. Improved Performance

- Uses **Microsoft's global edge network** to minimize latency.
- Caches static content at the edge to reduce backend server load.

2. High Availability & Scalability

- Provides automatic **failover** between different backend regions.
- Scales to handle **spikes in traffic** without manual intervention.

3. Enhanced Security

- Integrated **DDoS protection** and **Web Application Firewall (WAF)**.
- Supports **end-to-end encryption** and security policies.

4. Flexible Traffic Management

- Customizable rules for **geographic-based routing**, **latency-based routing**, and **priority-based routing**.
- Supports multiple routing methods for better traffic distribution.

5. Cost Optimization

- Reduces **bandwidth costs** by offloading static content at the edge.
- Lowers compute costs by reducing backend processing load.

6. Seamless Multi-Cloud and Hybrid Integration

- Works with **Azure, on-premises, and third-party cloud providers**.
- Enhances resilience for mission-critical applications.

Azure Front Door vs. Azure Traffic Manager vs. Azure CDN

Feature	Azure Front Door	Azure Traffic Manager	Azure CDN
Layer	Layer 7 (Application)	Layer 4 (DNS-based)	Layer 7 (Application)
Use Case	Web traffic routing & security	Global load balancing	Content caching
Security Features	WAF, DDoS protection	None	DDoS protection

Feature	Azure Front Door	Azure Traffic Manager	Azure CDN
Performance Optimization	Yes (caching, compression, TLS offload)	No	Yes (caching, acceleration)
Traffic Routing	URL-based, geo-based, latency-based	DNS-based failover	Edge caching

Conclusion

Azure Front Door is a powerful solution for **optimizing application performance, security, and reliability**. It is ideal for **high-traffic global applications, e-commerce platforms, SaaS applications, and APIs** that require low latency, intelligent routing, and protection against cyber threats.

The process involves setting up Azure Front Door to optimize traffic routing between two web apps hosted in different regions. First, two web apps are created in North Europe and the UK South. Then, in Azure, a Front Door profile is created with an endpoint and a route. An origin group is configured with both web apps, using HTTPS as the protocol. After deployment, accessing the Front Door endpoint directs traffic to the web app with the lowest latency. The end goal is to improve performance, availability, and load balancing across multiple regions for a seamless user experience.

😊 To begin with the Lab

1. To use Azure Front Door, you need to create two web apps in two different regions

Name ↑↓	Type ↑↓	Resource group ↑↓	Location ↑↓	Subscription ↑↓
ASP-NewRG-a36b	App Service plan	NewRG	UK South	MSDN Platforms Subscription
ASP-NewRG-b1c6	App Service plan	NewRG	North Europe	MSDN Platforms Subscription
northEuropeWebApp	App Service	NewRG	North Europe	MSDN Platforms Subscription
UKsouthWebApp12	App Service	NewRG	UK South	MSDN Platforms Subscription

2. Below you can see that we have two web apps running. Now if you don't know how to create web apps then you can follow the lab for traffic manager.

< > ↻ northeuropewebapp-gbbqhgcsf9fcd5ca.northeurope-01.azurewebsites.net

this the default page for north europe web application

this the deafult page for UK south web application

3. From the marketplace search for front door and CDN profiles. Click on Create.

Front Door and CDN profiles ...

Microsoft



Front Door and CDN profiles [Add to Favorites](#)

Microsoft | Azure Service

★ 2.9 (10 ratings)

Plan

Front Door and CDN profiles



Create

4. Choose Azure Front door then custom create. Click on continue.

Compare offerings ...

Microsoft Azure

Choose between Azure Front Door and other offerings.

Azure Front Door ☒

Azure Front Door is a secure cloud CDN which provides static and dynamic content acceleration, global load balancing and protection of your apps, APIs and websites with intelligent threat protection.

Explore other offerings ☐

See offerings for our Azure Front Door (classic) and Azure CDN Standard from Microsoft (classic), along with our partner offerings.

Choose between Azure Front Door options

Quick create ☐

Get started with a simplified web application deployment using default settings.

Define one endpoint with one origin and one WAF policy to get your Front Door up and running quickly.

Configure advanced settings and add endpoints as your needs evolve.

Custom create ☒

Leverage powerful configuration options to deploy a custom solution.

Design an endpoint with multiple domains and origin groups. Define routes to connect them, and add WAF policies to protect them.

Add endpoints to scale your deployment as your needs evolve.

[Continue to create a Front Door](#)

5. Choose your resource group and give it a unique name.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *	MSDN Platforms Subscription
Resource group *	NewRG
	Create new
Resource group location	North Europe

Profile details

Name *	frontdoor121
Tier * ⓘ	☒ Standard Content delivery optimized
	☐ Premium Security optimized

6. Then in the endpoint tab, click on add an endpoint.

*Basics Secrets ***Endpoint** Tags Review + create

Add an endpoint to get started. Add domains and create origin groups, then define routes to connect your domains and origin groups. Add security rules to complete your Front Door configuration. Later, you will be able to add endpoints and reuse your domains, origin groups, rule sets and WAF policies across endpoints in your Front Door profile. [Learn more](#)



No endpoints have been added yet

Try adding an endpoint to get started. [Learn more](#)

[Add an endpoint](#)

7. Give it a name and click on add.

Add an endpoint



Microsoft Azure

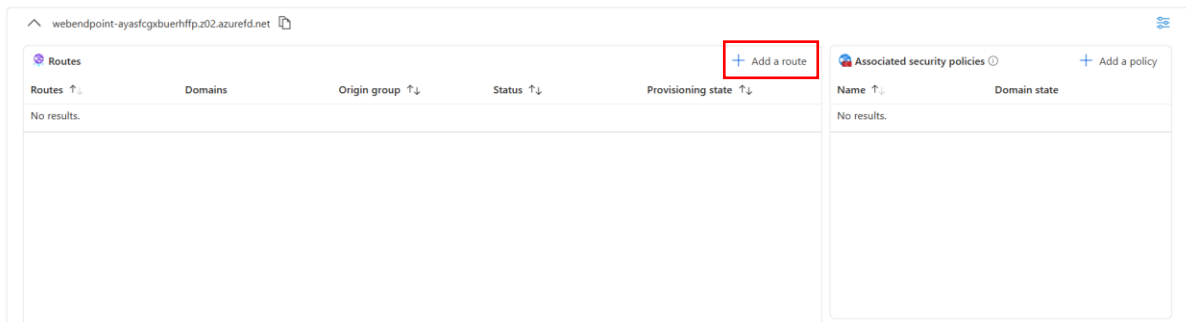
The endpoint name specifies a desired subdomain on Front Door's default domain (i.e. .z01.azurefd.net) to route traffic from that host via Front Door. You can optionally onboard custom domains as well. [Learn more](#)

Endpoint name *	webendpoint
Endpoint hostname	webendpoint-ayasfcgxbuerhffp.z02.azurefd.net
Status	☒ Enable this endpoint

8. After adding the endpoint, you need to add a route.

* Basics Secrets * **Endpoint** Tags Review + create

Add an endpoint to get started. Add domains and create origin groups, then define routes to connect your domains and origin groups. Add security rules to complete your Front Door configuration. Later, you will be able to add endpoints and reuse your domains, origin groups, rule sets and WAF policies across endpoints in your Front Door profile. [Learn more](#)



9. Here you need to give a name then you need to add a new origin group.

Add a route

Microsoft Azure

A route maps your domains and matching URL path patterns to a specific origin group. [Learn more](#)

Name *	webroute1
Endpoint	webendpoint-ayasfcgxbuerhffp.z02.azurefd.net
Domains	webendpoint-ayasfcgxbuerhffp.z02.azurefd.net
Patterns to match ⓘ	/ /path
Accepted protocols *	HTTP and HTTPS
Redirect	☒ Redirect all traffic to use HTTPS
Origin group	 Add a new origin group
Origin path	

10. Give your origin group a name and click on add an origin.

Add an origin group

Microsoft Azure



An origin group is a set of origins to which Front Door load balances your client requests. [Learn more](#)

Name *

web-origin



Origins

Origins are the application servers where Front Door will route your client requests. Utilize any publically accessible application server, including App Service, Traffic Manager, Private Link, and many others. [Learn more](#)

+ Add an origin

Origin host name

Status

Priority

Weight

Session affinity



Enable session affinity

11. We are going to add two origins, both of which are for the web apps.

Add an origin

Microsoft Azure



Origins are your application servers. Front Door will route your client requests to origins, based on the type, ports, priority, and weight you specify here. [Learn more](#)

[Go back to origin group](#)

Name *

uksouth-webapp



Origin type *

App services



Host name *

uksouthwebapp12-bvajbgh3hbcuh8ff.uksouth-01.azurewebsites.net



Origin host header

uksouthwebapp12-bvajbgh3hbcuh8ff.uksouth-01.azurewebsites.net



Certificate subject name validation ⓘ



Enable the validation



This validation is required if private link is enabled. [Learn more](#)

HTTP port *

80

HTTPS port *

443

Priority * ⓘ

1

Weight * ⓘ

1000

Status



Enable this origin

Add

Cancel

Close

Add an origin

Microsoft Azure



Origins are your application servers. Front Door will route your client requests to origins, based on the type, ports, priority, and weight you specify here. [Learn more](#)

[Go back to origin group](#)

Name *	northeurope-webapp
Origin type *	App services
Host name *	northeuropewebapp-gbbqhgcsf9fcd5ca.northeurope-01.azurewebsites.net
Origin host header	northeuropewebapp-gbbqhgcsf9fcd5ca.northeurope-01.azurewebsites.net
Certificate subject name validation ⓘ	☒ Enable the validation <i>This validation is required if private link is enabled. Learn more</i>
HTTP port *	80
HTTPS port *	443
Priority * ⓘ	1
Weight * ⓘ	1000
Status	☒ Enable this origin

Add

Cancel

Close

12. After adding the origins in the origin group choose the protocol as HTTPS. Click on add.

Origin host name	Status	Priority	Weight	
uksouthwebapp12-bvajbgh3hbcuh8ff.uksouth-01.azurewebsites.net	✓ Enabled	1	1000	...
northeuropewebapp-gbbqhgcsf9fcd5ca.northeurope-01.azurewebsites.net	✓ Enabled	1	1000	...

Session affinity ☐ Enable session affinity

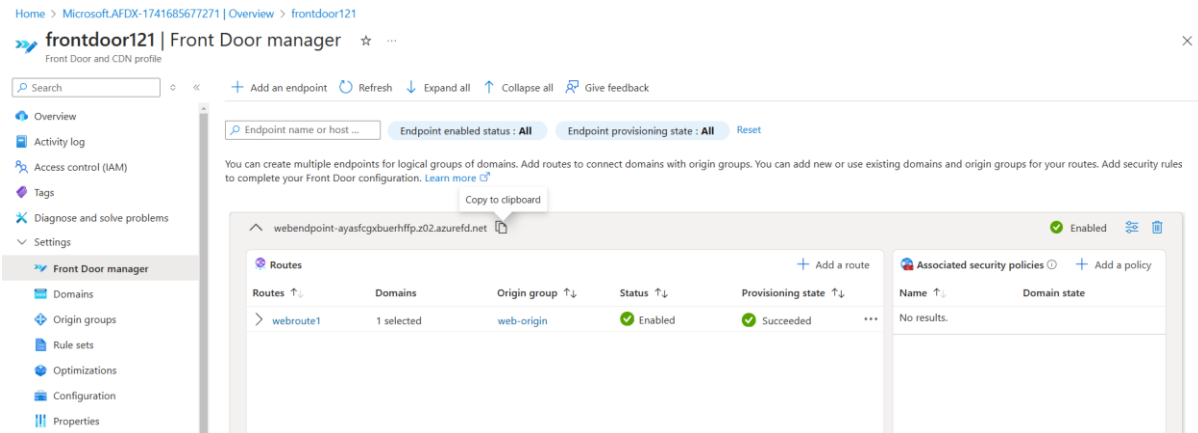
Health probes

If enabled, Front Door will send periodic requests to each of your origins to determine their proximity and health for load balancing purposes. [Learn more](#)

Status ⓘ	☒ Enable health probes
Path *	/
Protocol * ⓘ	☐ HTTP ☒ HTTPS
Probe method *	HEAD
Interval (in seconds) * ⓘ	100

seconds

13. Once your endpoint is settled then move to the review page and create your service.
14. Below you can see that our front door is live now and if go to the front door manager we can see the status of our web origin enables.
15. Now we need to copy the endpoint and paste it into a new tab.



16. Here you can see that we got the web app from the UK South location.
17. See, we had put two web apps as the origin, one in North Europe, the other in the UK South. Both of them had a priority of 1, and both of them had a weightage of 1000. Front Door is used to ensure that your request goes on to a web endpoint that has the lowest latency.



this the default page for UK south web application